United International **University (UIU)**
Dept. of Computer Science & Engineering (CSE)

Mid Exam Fall 2025

CSE 2233/CSI 233: Theory of Computation/Theory of Computing

Total Marks: **30**   Duration: 1 Hour 30 Minutes

Answer all questions. Figures are in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

---

1. Design DFAs that accepts the following languages: 3x3

a) L = { w | w **starts with** 101, **contains** 010 as substring and **ends with** 11 } where, ∑ ={0,1}
Example : 10101101011011, 101011011

b) L = { w | w **starts with** odd number of b and **total length** of the string is even} | ∑ = {a,b},
Example : bbba, babbaa

c) L = { $p^i q^j$ | **i** > **1**, **j** > **0** and **i+j** is an **even** number } where ∑ = { p,q }
Example: ppqq, pppq

2. **a)** Design NFAs that accept the following languages: 3x3

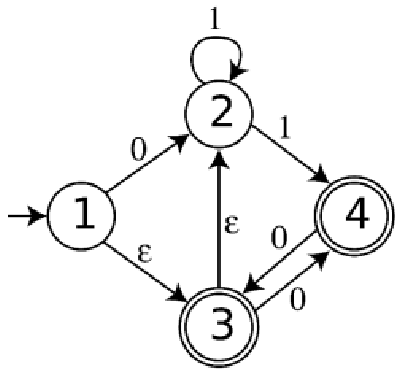
i) L = { w | w **starts with** "xyx", **contains** "yxz" or "xzy" and **ends with** "zyy" or "xxy"},
∑ = { x,y,z }, Example : xyxxzyxxy, xyxzyzyy

ii) L = { w | w **starts and ends** with **same symbol**, total length of the string is a **multiple of 3** if the string **starts with** x **or** total length of the string is a **multiple of 2** if the string **starts with** y or z }, ∑ = { x,y,z }, Example : xyzxyx, yxzy, zyxz

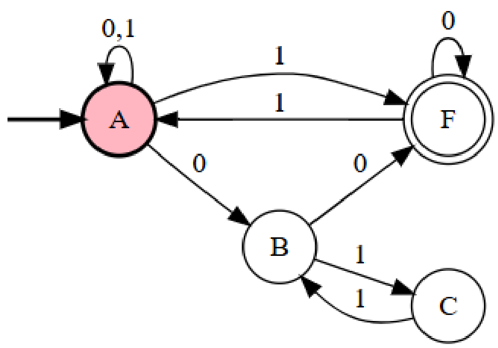
b) Design ε-NFAs that accept the following language

L = { w | w **starts with** "yz" or "xz", **contains** "zxy" as a substring, and **ends with** "xyx" or "zyz"}, ∑ = { x,y,z,ε }, Example : yzεzxyεxyx

3. Consider the following NFA, and show with the help of NFA-tree whether the string "**00110**" is accepted or not, $\sum = \{0,1\}$ 3



4. Convert the NFA to equivalent DFA. Show only the transition table (No state diagram needed). 3
Here, $\sum = \{0,1\}$:



5. Design Regular Expression for the following languages 2x3

a) L = { w | w contains at least two 0's but not two consecutive 0's }, $\sum = \{0,1\}$

b) L = { w | w does not contain xxx as substring }, $\sum = \{x,y\}$

c) L = { w | w starts with pq, contains ppq as a substring, and ends with qpq }, $\sum = \{p,q\}$